

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
INFORMATION TECHNOLOGY PROCUREMENT OFFICE (ITPO)
JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
(JOFOC)

For MathWorks Software and Support - 80TECH26QA027

1. Federal Acquisition Regulation (FAR) 6.104-1(a)(1) – Identification of the agency and the contracting activity, and specific identification of the document as a “Justification for other than full and open competition.”

This document is a justification for other than full and open competition prepared by National Aeronautics and Space Administration (NASA) Information Technology Procurement Office (ITPO).

The procuring agency is NASA, and the contracting activity is the ITPO.

2. FAR 6.104-1(a)(2) – The nature and/or description of the action being approved:

This justification provides the rationale for contracting by other than full and open competition to award a sole-source Single-Award Blanket Purchase Agreement (BPA) for The MathWorks, Inc. (MathWorks) software and services. NASA is seeking to procure MathWorks software and services through a Single-Award BPA to provide MathWorks software products and services including MATLAB®, Simulink®, and Polyspace™ Product Families and MATLAB® Distributed, Computing Server and MATLAB® Production Server associated support for license usage. This agreement will provide MathWorks products and services on an as-needed basis throughout the five-year duration of the agreement.

3. FAR 6.104-1(a)(3) – A description of the supplies or services required, to meet the Agency’s needs (including the estimated value):

NASA requires the establishment of a BPA with MathWorks to provide proprietary mathematical computing software licenses, software maintenance, and associated technical support necessary to support ongoing and future NASA research, engineering, and mission operations activities.

The BPA will provide access to MATLAB, Simulink, and associated toolboxes and products required for modeling, simulation, data analysis, algorithm development, system design, verification and validation, and flight and ground systems analysis. These products are used across multiple NASA Centers and programs in support of aeronautics research, space exploration, science missions, and mission assurance activities.

The BPA will enable NASA to procure:

- New and renewal licenses for MATLAB, Simulink, and related MathWorks toolboxes.
- Software maintenance and updates, including access to new releases, patches, and security updates.
- Vendor-provided technical support necessary to maintain operation continuity and mission readiness.

Establishing a BPA with MathWorks ensures uninterrupted access to mission-critical software that is foundational to NASA's engineering, research, and mission assurance activities. The BPA provides the flexibility to meet diverse and evolving program needs while maintain standardization, interoperability, and continuity with existing analytical assets. This approach minimizes administrative burden, reduces procurement lead times, and mitigates technical and schedule risk across the Agency. Additionally, establishing a single BPA vehicle directly with MathWorks allows NASA to leverage volume efficiencies, standardize terms and conditions, and streamline ordering across Centers and programs.

The period of performance will be for a five (5) year ordering period. The estimated potential value of the BPA is [REDACTED]. The BPA will be operating under FAR 12.201-1 procedures, with the value of each order limited to [REDACTED].

4. FAR 6.104-1(a)(4) – An identification of the statutory authority permitting other than full and open competition:

The statutory authority permitting other than full and open competition for acquisitions using the procedures in FAR subpart 12.102(b), Table 12-1 is 41 U.S.C. 1901.

5. FAR 6.104-1(a)(5) – A demonstration that the proposed contractor's unique qualifications or the nature of the acquisition requires use of the authority cited:

The rationale supporting the use of 41 U.S.C. 1901 is as described below.

MathWorks is the sole developer, owner, and authorized distributor of MATLAB, Simulink, and associated proprietary toolboxes. These products are protected by intellectual property rights and licensing restrictions that preclude third-party development, distribution, or support of equivalent products that are fully compatible with existing MATLAB and Simulink environments. As a result, only MathWorks can provide the required software licenses, maintenance, updates, and vendor-authorized technical support.

MATLAB and Simulink are deeply embedded in NASA's technical infrastructure and are extensively used across multiple Centers and programs to support modeling, simulation, data analysis, algorithm development, system design, and verification and validation activities. NASA engineers and researchers have developed a substantial body of mission-critical models, scripts, and simulation frameworks using these tools. These assets are integral to ongoing and future aeronautics research, spaceflight programs, science missions, and mission assurance activities.

MATLAB and Simulink are core tools supporting NASA's digital engineering and model-based systems engineering (MBSE) initiatives. These tools enable executable models, traceability between requirements and design, and early system validation. Continued access to MathWorks products is necessary to sustain ongoing digital transformation efforts and ensure consistency across program lifecycles.

No other commercial or open-source software products provide a fully equivalent combination of functionality, toolchain integration, and compatibility with NASA's existing MATLAB and Simulink-based models and workflows. Substitution with alternative products would require extensive redevelopment, revalidation, and recertification of existing models and analyses, resulting in substantial duplication of cost, significant schedule delays, and increased technical and mission risk.

Additionally, continued access to vendor-provided software maintenance and technical support from MathWorks is required to ensure software reliability, security, and compatibility with evolving computing environments. Only MathWorks can provide authorized updates, patches, and defect resolutions for its proprietary software, which are necessary to maintain operational continuity and mission readiness.

For these reasons, MathWorks is the only responsible source capable of meeting NASA's requirements for the supplies and services described in this justification.

6. FAR 6.104-1(a)(6) – A description of the efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by part 5 and, if not, which exception under part 5 applies:

A notice to the Government Point of Entry (GPE) website (Sam.gov) was published on 2/25/2026, in accordance with FAR part 5. This posting informed potential sources of NASA's intent to award this sole-source contract to MathWorks. The results of this synopsis are summarized in Section 10 below.

7. FAR 6.104-1(a)(7) – A determination by the contracting officer that the anticipated cost to the Government will be fair and reasonable:

The Contracting Officer's signature on this document indicates that the Contracting Officer has determined that the anticipated cost to the government will be fair and reasonable. The Contractor shall be required to submit a quote to be evaluated and negotiated by the Government. Prior to execution of the contractual instrument, a price analysis will be performed to ensure that the final agreed-to price for the contract action is fair and reasonable. In accordance with FAR 12.204(a), price reasonableness will be determined through a comparison of proposed prices, similar requirements, historical and current market prices to determine acceptability and reasonableness.

8. FAR 6.104-1(a)(8) – Description of the market research conducted, and the results, or a statement of the reason market research was not conducted (e.g., urgent, and compelling requirement):

Market research indicates that no commercial or open-source alternatives are capable of fully replicating MathWorks MATLAB and Simulink’s integrated functionality, toolchain support, and compatibility required for NASA’s model-based workflows. Although open-source solutions and similar tools provide certain features within specific domains, they lack the seamless integration, automatic code generation, and direct compatibility with existing models necessary to meet NASA’s requirements. Likewise, commercial offerings may serve as complements, but do not function as standalone replacements and often necessitate additional migration or interface layers. Accordingly, only MathWorks MATLAB and Simulink are positioned to deliver the comprehensive graphical modeling, real-time execution, and complete toolchain integration essential for NASA’s engineering and testing processes.

Market research indicates that while several commercial and open source software platforms provide numerical computing or model based design (MBD) capabilities similar to MATLAB and Simulink, they do not meet the full suite of characteristics that NASA requires for flight critical development, verification, and systems engineering workflows. Tools such as GNU Octave, Scilab/Xcos, Python ecosystems, Julia, and Modelica based environments (e.g., OpenModelica or Altair Activate) can replicate portions of MATLAB’s numerical analysis or block diagram modeling features. However, these platforms lack turnkey tool qualification artifacts, aerospace specific libraries, and the integrated digital engineering toolchain that MathWorks provides. As a result, they do not provide an equivalent capability for NASA programs that must adhere to rigorous standards such as DO 178C, DO 254, ARP4754B, and ARP4761A.

Research also shows that several commercial system simulation tools—such as Siemens Simcenter Amesim, COMSOL Multiphysics, and Wolfram System Modeler—can perform advanced multiphysics or system dynamics modeling comparable to portions of the Simulink environment. These tools are strong in their respective domains but typically lack the end to end workflows used within NASA for flight software development, including automatic code generation integrated with verification frameworks, real time hardware in the loop (HIL) execution, and agency preferred requirements traceability. Unlike MathWorks, these alternative vendors do not provide comprehensive DO 178C/DO 254 tool qualification packages or aviation focused design toolboxes that directly support NASA’s GN&C, mission analysis, and avionics engineering practices.

Additionally, NASA programs have a long history of deploying MATLAB/Simulink across mission profiles, including GN&C development for Orion, small satellite platforms, robotics, and UAV research—which gives the MathWorks ecosystem substantial flight heritage that competing tools do not currently match. Even emerging software, such as Rust based model based design environments, shows potential for future aerospace applications but does not yet provide the same maturity, ecosystem depth, or certification artifacts needed for operational mission use. Based on the surveyed market, MathWorks remains the only vendor

offering an integrated, certification ready, aerospace oriented MBD environment with proven NASA adoption.

Market research also indicates that MathWorks is the only source capable of meeting NASA's requirements for a multi-year agency-wide agreement for MathWorks software products and services. MathWorks' pricing structure requires NASA to establish a five-year ordering period to obtain discounted pricing below standard commercial rates. In its Sole Source Letter, MathWorks confirmed that it does not offer fixed pricing for future years and that all transactional purchases must be based on its then-current published commercial price list. The proposed BPA structure provides NASA with an upfront ten percent discount off commercial pricing. Without a consolidated multi-year agreement directly with MathWorks, NASA would be required to purchase licenses at higher, fluctuating commercial rates, resulting in significantly increased cost to the Government.

MathWorks confirmed that it is the sole authorized provider of software delivery, technical support, maintenance, and updates for all MathWorks products in the United States and Canada, and it does not permit any third-party organizations to perform these functions. As part of NASA's market research, SEWP was reviewed and several Value Added Resellers were identified; however, MathWorks has stated that resellers are not acceptable for this requirement. In its Sole Source Letter, MathWorks confirmed that it is the sole manufacturer, publisher and owner of MathWorks software products and the sole provider of any updates or technical support for these products in the United States and Canada. MathWorks also confirmed that it does not offer nor does it authorize resellers in North America to provide multi-year pricing to commercial accounts.

Because MathWorks updates its price lists multiple times per year and does not offer fixed future-year pricing, only MathWorks can provide the pricing structure required for NASA's transactional BPA purchases. Additionally, MathWorks has successfully supported NASA under IDIQ 80NSSC21D0001 and will continue to maintain the staff and resources necessary to support a new NASA-wide BPA. Procuring directly from MathWorks ensures uninterrupted service from the sole manufacturer, avoids delays in order processing, maintains continuity in account and records management, and provides access to dedicated account support. Based on this market research, no other vendor can meet NASA's technical, pricing, or support requirements for this procurement.

9. FAR 6.104-1(a)(9) – Any other facts supporting the use of other than full and open competition:

None

10. FAR 6.104-1(a)(10) – A listing of the sources, if any, that expressed an interest in writing in the acquisition:

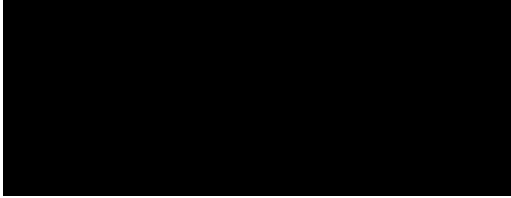
A notice of NASA's intent to award this sole-source action was synopsisized on the GPE website (Sam.gov) per FAR part 5 (See Section 6 above) on 2/25/2026 and closed on 3/12/2026 and no companies responded.

11. **FAR 6.104-1(a)(11) – A statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required:**

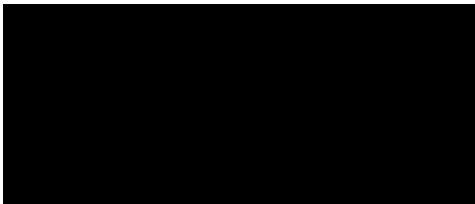
The Agency will continue to examine the market in the future for alternative solutions or new sources before executing any subsequent acquisitions for the same requirements.

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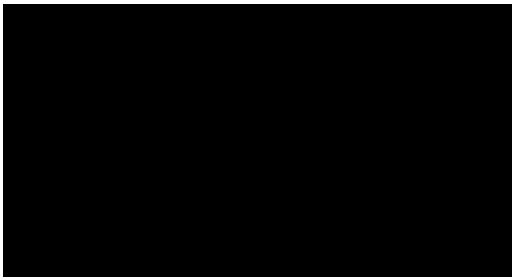
I certify that the facts presented in this justification are accurate and complete.



I hereby certify that the above justification is complete and accurate to the best of my knowledge and belief.



CONCURRENCES:



APPROVAL:

